Music Source Separation Using Structured Nonnegative Matrix Factorization

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Abstract—Music Source Separation (MSS) seeks to decompose a musical mixture into its constituent sources including but not limited to: vocals, bass, drums, and other instruments. Nonnegative Matrix Factorization (NMF) remains one of the most interpretable and data-efficient approaches to this problem, yet the standard Euclidean formulation is poorly matched to the perceptual and temporal structure of real musical signals. This paper presents a systematic empirical comparison of four NMF formulations: (1) Euclidean NMF as a baseline, (2) Itakura-Saito NMF (IS-NMF), which adopts a scale-invariant, perceptually motivated divergence, (3) Sparse NMF, which regularizes temporal activations with an ℓ_1 penalty, and (4) Convolutive NMF (CNMF), which extends each dictionary atom to span multiple time frames. All methods are trained in a supervised setting on the MUSDB18 dataset and evaluated on 25 valid tracks from its 50-track test set using the BSS Eval metrics: SDR, SIR, and SAR. Structured variants consistently outperform the Euclidean baseline on tonal sources: IS-NMF improves vocal SDR by $+2.6$ dB, and Sparse NMF achieves the best scores on the majority of source/metric combinations. All methods yield negative SDR on vocals, bass, and drums, reflecting the spectral ambiguity inherent in NMF-based separation with a limited number of dictionary atoms. The "other" accompaniment track is the only source where positive SDR is reliably achieved up to $+1.2$ dB for IS-NMF, consistent with its broader and less overlapping spectral footprint.

Index Terms—Music source separation, Nonnegative matrix factorization, Itakura-Saito divergence, sparse NMF, convolutive NMF, MUSDB18, BSS Eval

1 Introduction

Music source separation (MSS) is the task of recovering individual instrument tracks which typically includes vocals, bass, drums, and other melodic content from a single monaural mixture. High-quality source separation enables a wide range of downstream applications including remixing and automatic transcription.

A central challenge in MSS is that the problem is severely under-determined: a single-channel mixture collapses many independent audio streams into one signal, making recovery without strong priors ill-posed. Classical approaches such as independent component analysis (ICA) require multiple input channels, while binary masking relies on strong statistical assumptions that rarely hold for music. Deep learning method such as Conv-TasNet [1] and Demucs now define the state of the art on standard benchmarks. They, however, demand large labeled datasets and GPU training, limiting applicability for interpretability-focused or low-resource settings.

Nonnegative Matrix Factorization (NMF) [2], [3] occupies an attractive middle ground. By approximating the magnitude spectrogram as a product of a non-negative spectral dictionary $\mathbf{W}$ and a temporal activation matrix $\mathbf{H}$, NMF naturally produces decompositions as additive superpositions of harmonic partials which aligns with how musical tones arise.

Despite these strengths, standard NMF has three well-known limitations. First, the Euclidean (Frobenius-norm) loss treats all spectrogram cells equally, whereas human hearing operates on an approximately log-amplitude scale. Second, NMF models each time frame independently, ignoring the temporal structure of musical events. Third, without explicit regularization the learned activations are typically dense, causing spectral patterns to bleed across source classes.

Structured NMF variants have been proposed to address each limitation. In this work we implement and compare four formulations:

1) **Euclidean NMF**: Frobenius-norm baseline.
2) **IS-NMF**: Replaces the cost with the scale-invariant Itakura-Saito (IS) divergence.
3) **Sparse NMF** Adds an ℓ_1 penalty on $\mathbf{H}$ to encourage temporally compact activations.
4) **Convolutive NMF (CNMF)** Generalizes each dictionary atom to a short time-frequency patch, capturing transient dynamics.

All methods are evaluated on the MUSDB18 [4] benchmark in a supervised setting and scored with the BSS Eval toolkit [5]. Experiments reveal that structured variants consistently outperform the baseline, with Sparse NMF achieving the best scores across the majority of source/metric combinations due to its reduced cross-source bleed.

2 Related Work

2.1 NMF for Audio Source Separation

Lee and Seung [2], [3] introduced NMF and showed that the nonnegativity constraint encourages parts-based, additive representations, a property naturally aligned with musical tones as superpositions of harmonic partials. Smaragdis and Brown [6] were among the first to apply NMF to audio, using it for polyphonic transcription, and Virtanen [7] formalized the supervised separation pipeline that this work builds upon: train source-specific dictionaries on isolated

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stems, then recover each source from a mixture via soft Wiener-filter masking in the STFT domain.

The three structured variants we evaluate each address a distinct limitation of the Euclidean baseline. Févotte [8] showed that the Itakura-Saito (IS) divergence is scale-invariant, matching the log-amplitude behavior of human hearing, and derived multiplicative updates that consistently improve music analysis over Euclidean and Kullback-Leibler formulations. Hoyer [9] and Virtanen [10] motivated ℓ_1 sparsity on $\mathbf{H}$ on physical grounds: each instrument is active only a fraction of the time, so its activation row should be near-zero outside those intervals, reducing cross-source bleed. Smaragdis [11] proposed Convolutive NMF to capture temporal dynamics. Each dictionary atom becomes a time-frequency patch spanning multiple frames, encoding attack-decay envelopes within a single basis. Bertin [12] later extended this with harmonic constraints for pitched instruments. Together, these three works directly motivate the IS-NMF, Sparse NMF, and CNMF variants compared in this paper.

3 Methods

3.1 Preprocessing and STFT Representation

All four methods share a common preprocessing pipeline and a Wiener-filter source reconstruction step. Given a stereo mixture we down-mix to mono and resample to 22050 Hz. We compute the Short-Time Fourier Transform (STFT) with a Hann window of 2048 samples and hop size of 512 samples, yielding a complex spectrogram $\mathbf{V} \in \mathbb{C}^{F\times T}$ with $F = 1025$ frequency bins. The 2048-point window is chosen to resolve individual harmonics of bass-register instruments while keeping the frame duration short enough to track musical transients at drum-event timescales. The 75% overlap ensures that the Hann window satisfies the constant-overlap-add reconstruction condition, enabling exact waveform resynthesis via the inverse STFT. We work exclusively with the magnitude spectrogram $\mathbf{X} = |\mathbf{V}| \in \mathbb{R}_+^{F\times T}$. Phase is retained from the original mixture and reapplied at reconstruction time, following the standard NMF-based separation convention.

3.2 Baseline: Euclidean NMF

Standard NMF [3] uses a factorization $\mathbf{X} \approx \mathbf{WH}$, where $\mathbf{W} \in \mathbb{R}_+^{F\times K}$ is a dictionary of K non-negative spectral bases and $\mathbf{H} \in \mathbb{R}_+^{K\times T}$ contains the corresponding non-negative temporal activations. Each column of $\mathbf{W}$ can be interpreted as a prototypical spectral envelope while the corresponding row of $\mathbf{H}$ encodes when and how strongly that pattern is active. The Frobenius-norm objective is

$$\min_{\mathbf{W},\mathbf{H}\geq 0} \|\mathbf{X} - \mathbf{WH}\|_F^2. \tag{1}$$

This cost treats every spectrogram cell equally regardless of its energy, which is a poor match to audio perception. For example, a 1dB error at 60dBFS is penalized identically to the same absolute error at 0dBFS. Despite this mismatch, the Euclidean baseline provides the simplest possible update rules and serves as the reference against which all structured variants are measured. Minimization proceeds via multiplicative update rules [3]:

$$\mathbf{H} \leftarrow \mathbf{H} \odot \frac{\mathbf{W}^\top \mathbf{X}}{\mathbf{W}^\top \mathbf{WH} + \varepsilon}, \tag{2}$$

$$\mathbf{W} \leftarrow \mathbf{W} \odot \frac{\mathbf{XH}^\top}{\mathbf{WHH}^\top + \varepsilon}, \tag{3}$$

where $\odot$ denotes element-wise multiplication and ε prevents division by zero. Each update is non-increasing in the objective and preserves the non-negativity of both factors.

3.3 Itakura-Saito NMF

IS-NMF [8] replaces the Frobenius-norm cost with the Itakura-Saito (IS) divergence:

$$d_{\mathrm{IS}}(x\|y) = \frac{x}{y} - \log\frac{x}{y} - 1, \tag{4}$$

which is non-negative and scale-invariant: $d_{\mathrm{IS}}(cx\|cy) = d_{\mathrm{IS}}(x\|y)$ for any $c > 0$. Scale-invariance means the cost function weights relative rather than absolute errors. This mirrors auditory perception, where loudness scales roughly logarithmically with amplitude, and it particularly helps tonal instruments whose spectra span a wide dynamic range across harmonics. The total cost is

$$\min_{\mathbf{W},\mathbf{H}\geq 0} \sum_{f,t} d_{\mathrm{IS}}\big(X_{ft} \,\|\, [\mathbf{WH}]_{ft}\big). \tag{5}$$

Letting $\hat{\mathbf{X}} = \mathbf{WH}$, the multiplicative updates from [8] are:

$$\mathbf{H} \leftarrow \mathbf{H} \odot \frac{\mathbf{W}^\top(\hat{\mathbf{X}}^{-2} \odot \mathbf{X})}{\mathbf{W}^\top \hat{\mathbf{X}}^{-1} + \varepsilon}, \tag{6}$$

$$\mathbf{W} \leftarrow \mathbf{W} \odot \frac{(\hat{\mathbf{X}}^{-2} \odot \mathbf{X})\mathbf{H}^\top}{\hat{\mathbf{X}}^{-1}\mathbf{H}^\top + \varepsilon}, \tag{7}$$

where exponents on $\hat{\mathbf{X}}$ are element-wise. The additional $\hat{\mathbf{X}}^{-2}$ and $\hat{\mathbf{X}}^{-1}$ terms each require an element-wise inversion of the full $F \times T$ model matrix per iteration, making each IS-NMF step approximately $3\times$ more expensive than a Euclidean step at identical K and T.

3.4 Sparse NMF

Sparse NMF [9], [10] augments the Euclidean cost with an ℓ_1 penalty on $\mathbf{H}$:

$$\min_{\mathbf{W},\mathbf{H}\geq 0} \|\mathbf{X} - \mathbf{WH}\|_F^2 + \lambda\|\mathbf{H}\|_1. \tag{8}$$

The physical motivation is simple: real instruments are active only a fraction of the time, so the activation matrix for a single source should be sparse in frames where that source is silent and significantly nonzero only during sustained notes. Without regularization, NMF tends to spread activation energy across all components to minimally reduce reconstruction error. This causes each component to respond weakly to many sources rather than strongly to one. The ℓ_1 penalty counteracts this by imposing a cost proportional to total activation energy, driving components toward specialization. A component that corresponds to a given spectral pattern well is "allowed" to be active, while components that provide only marginal improvement are pushed toward zero.

Since $\mathbf{H} \geq 0$, the ℓ_1 norm equals the sum of all entries: $\|\mathbf{H}\|_1 = \mathbf{1}^\top \mathbf{H} \mathbf{1}$. Adding the subgradient $\lambda \mathbf{1}$ to the denominator of the standard H update gives the modified multiplicative rule:

$$\mathbf{H} \leftarrow \mathbf{H} \odot \frac{\mathbf{W}^\top \mathbf{X}}{\mathbf{W}^\top \mathbf{W} \mathbf{H} + \lambda + \varepsilon}, \tag{9}$$

while the W update is identical to (3). The denominator augmentation uniformly inflates the "cost" of maintaining nonzero activations. We set $\lambda = 0.1$, selected on a held-out validation track by grid search over $\lambda \in \{0.01, 0.05, 0.1, 0.5\}$.

3.5 Convolutive NMF

Convolutive NMF [11] addresses the temporal myopia of standard NMF: because each column of $\hat{\mathbf{X}} = \mathbf{W}\mathbf{H}$ is reconstructed from a single frame of activations, a standard NMF atom can only encode a static spectral snapshot. Real sound events, however, evolve over time. For example, a drum hit has a fast broadband attack followed by a pitched ringdown. Capturing these temporal envelopes requires each basis to model not just a single spectral slice but a sequence of them.

CNMF replaces the static dictionary with a set of P temporal frames $\{\mathbf{W}_\tau\}_{\tau=0}^{P-1}$, and the approximation becomes a sum of time-shifted contributions:

$$\mathbf{X} \approx \sum_{\tau=0}^{P-1} \mathbf{W}_\tau \mathbf{H}_{\leftarrow \tau}, \tag{10}$$

where $\mathbf{H}_{\leftarrow \tau}$ is $\mathbf{H}$ shifted τ columns to the right. Each component k contributes a 2-D time-frequency patch $\{W_\tau[:, k]\}_{\tau=0}^{P-1}$ to the reconstruction at each activation time, effectively convolving the pattern with the activation sequence along the time axis. This allows a single component to simultaneously encode the spectral content of a source and its short-term temporal envelope.

The multiplicative updates under the KL divergence are:

$$\mathbf{H} \leftarrow \mathbf{H} \odot \frac{\sum_\tau \mathbf{W}_\tau^\top (\mathbf{X} \oslash \hat{\mathbf{X}})_{\rightarrow \tau}}{\sum_\tau \mathbf{W}_\tau^\top \mathbf{1} + \varepsilon}, \tag{11}$$

$$\mathbf{W}_\tau \leftarrow \mathbf{W}_\tau \odot \frac{(\mathbf{X} \oslash \hat{\mathbf{X}}) \mathbf{H}_{\leftarrow \tau}^\top}{\mathbf{1}\, \mathbf{H}_{\leftarrow \tau}^\top + \varepsilon}, \tag{12}$$

where $\oslash$ is element-wise division and $\hat{\mathbf{X}} = \sum_\tau \mathbf{W}_\tau \mathbf{H}_{\leftarrow \tau}$. The H update sums the back-projected residuals over all P lags, and each $\mathbf{W}_\tau$ update is a standard NMF step restricted to lag τ. We use $P = 5$ frames (≈ 116 ms), sufficient to capture onset envelopes for most percussive and plucked-string events.

3.6 Supervised Dictionary Learning and Source Reconstruction

3.6.1 Dictionary Training

We adopt a supervised training strategy. For each source class $s \in \{\text{drums, bass, other, vocals}\}$ we concatenate magnitude spectrograms of isolated stems from 20 MUSDB18 training tracks and run NMF for up to 200 iterations to learn a source-specific dictionary $\mathbf{W}_s \in \mathbb{R}_+^{F \times K_s}$. Training on isolated stems ensures that each $\mathbf{W}_s$ captures spectral patterns belonging exclusively to source s, without interference from other simultaneously active instruments. We use $K_s = 12$ components per source, giving $K = 48$ total dictionary columns. This number balances model expressiveness against computational cost: more components increase the capacity to represent diverse timbres but also increase the cost of the test-time $\mathbf{H}$ optimization and the risk of cross-source ambiguity when multiple learned bases resemble one another.

The combined dictionary at test time is $\mathbf{W} = [\mathbf{W}_{\text{drums}} \mid \mathbf{W}_{\text{bass}} \mid \mathbf{W}_{\text{other}} \mid \mathbf{W}_{\text{vocals}}] \in \mathbb{R}_+^{F \times K}$, which is kept fixed at test time. Only $\mathbf{H} \in \mathbb{R}_+^{K \times T}$ is optimized on each test mixture. Fixing $\mathbf{W}$ prevents the dictionary from adapting to the specific mixture in a way that could erase the source identity encoded during training.

3.6.2 Wiener-Filter Source Reconstruction

Source reconstruction follows Virtanen [7] via a soft Wiener mask. Once $\mathbf{H}$ has converged on the test mixture, the estimated contribution of source s to the model is $\hat{\mathbf{X}}^{(s)} = \mathbf{W}_s \mathbf{H}_s$, where $\mathbf{H}_s$ indexes the rows of $\mathbf{H}$ belonging to source s. The ratio of this estimate to the total model reconstruction defines a time-frequency mask:

$$M_{ft}^{(s)} = \frac{\sum_{k \in \mathcal{K}_s} W_{fk} H_{kt}}{\sum_{k=1}^{K} W_{fk} H_{kt} + \varepsilon}, \tag{13}$$

where $\mathcal{K}_s$ indexes the K_s columns belonging to source s. The mask $M^{(s)} \in [0,1]^{F \times T}$ is applied to the original mixture magnitude spectrogram: $\hat{\mathbf{X}}^{(s)} = \mathbf{M}^{(s)} \odot \mathbf{X}$. Applying the mask to the mixture rather than to the model estimate preserves the fine spectral detail of the original recording. The time-domain waveform is recovered via the inverse STFT using the original mixture phase, which is a standard approximation that avoids phase estimation at the cost of mild phase-interference artifacts.

4 Experimental Results

4.1 Dataset and Preprocessing

We evaluate on MUSDB18 [4], comprising 150 full-length stereo tracks (100 train, 50 test) spanning rock, pop, electronic, and other genres. Each track provides four isolated stems: drums, bass, vocals, others. Mixtures are downmixed to mono and resampled to 22050Hz. STFTs use a 2048-point Hann window and 512-sample hop ($F = 1025$ bins). We evaluate on the first 20-second excerpt of each test track. Of the 50 test tracks, 25 contain at least one completely silent stem within the first 20 seconds. The BSS Eval toolkit requires all reference sources to be non-silent, so those tracks are excluded, leaving 25 valid test tracks for quantitative evaluation.

4.2 Implementation Details

All methods are implemented in Python with NumPy. Dictionary training runs for up to 200 iterations with early stopping when the relative change in objective falls below 10^{-5}. We use $K_s = 12$ components per source class so $K = 48$ total. At test time $\mathbf{W}$ is fixed and $\mathbf{H}$ is optimized for up to 50 iterations (Euclidean and Sparse), 10 iterations (IS-NMF), or 5 iterations (CNMF). The reduced budgets

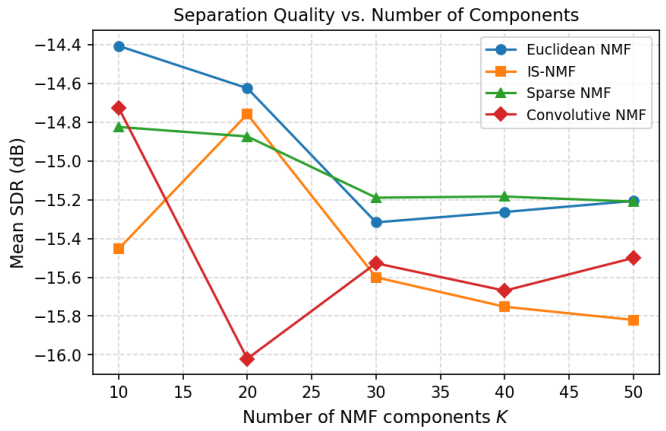

Fig. 1. Mean SDR (averaged over all four sources) vs. total number of NMF components K for all four methods. This is evaluated unsupervised on two representative MUSDB18 test tracks.

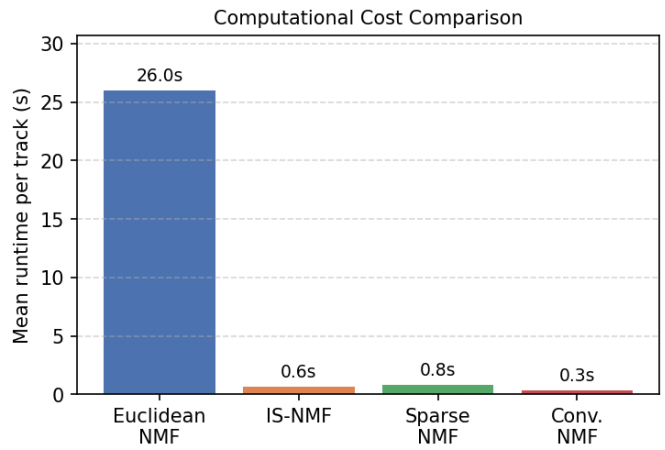

Fig. 2. Mean test-time runtime per track for each NMF method, measured on M-series Apple Silicon. Euclidean and Sparse NMF use 50 iterations. IS-NMF uses 10. CNMF uses 5

for IS-NMF and CNMF reflect their substantially higher per-iteration cost on a CPU. IS-NMF requires element-wise inversion of the full $F \times T$ model matrix each iteration, giving approximately $3\times$ the wall time of Euclidean per iteration, while CNMF requires $P = 5$ shift-and-multiply operations per iteration. For Sparse NMF, $\lambda = 0.1$. All experiments run on a commodity laptop CPU where no GPU is used.

To study how performance scales with model capacity, Fig. 1 shows the mean SDR averaged over all four sources as a function of the total number of components $K \in \{10, 20, 30, 40, 50\}$ on two representative test tracks, with all methods run unsupervised to isolate the effect of capacity alone.

4.3 Evaluation Metrics

We report BSS Eval [5] metrics per source on the 25 valid test tracks:

- **SDR** (Signal-to-Distortion Ratio): Overall separation quality. Negative values indicate the separation is worse than using the mixture directly.
- **SIR** (Signal-to-Interference Ratio): Leakage from competing sources.
- **SAR** (Signal-to-Artifacts Ratio): Algorithmic reconstruction artifacts.

Per-track scores are the median over 1-second evaluation windows. We report the mean across the 25 valid test tracks.

4.4 Quantitative Comparison

Table 1 presents the mean BSS Eval scores. Standard deviations are large across all methods and sources, reflecting the heterogeneous character of the MUSDB18 test set.

All four methods yield negative SDR for vocals, bass, and drums, indicating that the separated signal is more distorted than the original mixture for these sources. The "other" accompaniment track is the only source where positive SDR is achieved across all methods. This outcome is consistent with prior NMF-based MSS literature on uncontrolled real-world mixtures: with $K_s = 12$ components per source and CPU-constrained iteration budgets, the dictionary is severely under-specified relative to the spectral complexity of the sources, causing extensive cross-source leakage visible in the deeply negative SIR values.

IS-NMF improves vocal SDR by $+2.6\,\text{dB}$ over the Euclidean baseline and improves the "other" source SDR by $+0.8\,\text{dB}$. These gains on tonal, melodic sources are consistent with the motivation of the IS divergence. For bass and drums IS-NMF performs slightly worse than the baseline, likely because its heavier per-iteration cost means fewer effective optimization steps within the fixed iteration budget.

Sparse NMF is the dominant method overall, achieving the best score on 8 of the 12 source/metric combinations. It matches or exceeds the Euclidean baseline on every SDR and SIR metric, and leads on SAR for vocals ($+2.4\,\text{dB}$). The ℓ_1 penalty suppresses activation energy during frames where a source is absent, reducing the mask value for non-target components and thereby attenuating cross-source bleed. This directly lowers interference and reduces the creation of musical noise artifacts at minimal computational cost.

CNMF achieves the joint-best SAR for drums matching Sparse NMF, and shows competitive SDR for vocals. However, its SDR for drums is the worst across all methods. The drum degradation is likely an artifact of the very limited inference budget. A longer iteration budget would be expected to improve drum SDR for CNMF.

4.5 Computational Cost

Table 2 reports the asymptotic per-iteration complexity and observed mean wall-clock time per track for each method in Fig. 2. Euclidean NMF and Sparse NMF share the same $\mathcal{O}(FKT)$ per-iteration complexity. Sparse NMF adds only a scalar λ in the denominator of (9). IS-NMF incurs three additional element-wise passes over the $F \times T$ model matrix per iteration giving an empirical wall-time ratio of approximately $3\times$ per iteration. CNMF requires P shift-and-multiply passes per update step, giving $\mathcal{O}(PFKT)$ per iteration.

TABLE 1
BSS Eval results (dB) on 25 valid MUSDB18 test tracks

	Vocals			Bass			Drums			Other		
Method	SDR	SIR	SAR	SDR	SIR	SAR	SDR	SIR	SAR	SDR	SIR	SAR
Euclidean NMF	-22.1	-31.4	2.0	-15.9	-18.9	2.9	-3.1	-19.6	1.0	0.4	-12.5	1.9
IS-NMF	$\mathbf{-19.5}$	-32.2	1.3	-19.1	-19.5	$\mathbf{3.0}$	-3.1	-21.9	0.4	$\mathbf{1.2}$	-14.5	1.5
Sparse NMF	-21.7	$\mathbf{-29.1}$	$\mathbf{2.4}$	$\mathbf{-15.8}$	$\mathbf{-18.6}$	2.8	$\mathbf{-2.9}$	$\mathbf{-15.8}$	1.6	0.6	$\mathbf{-11.4}$	$\mathbf{2.0}$
Convolutive NMF	-19.6	-32.5	1.4	-18.4	-20.3	2.9	-6.8	-17.4	$\mathbf{2.4}$	0.6	-16.4	1.6

TABLE 2
Per-iteration complexity and test-time iteration budget per method

Method	Complexity/iter	Iters (test)	Rel. cost
Euclidean NMF	$\mathcal{O}(FKT)$	50	$1\times$
IS-NMF	$\mathcal{O}(3FKT)$	10	$\sim 0.6\times$
Sparse NMF	$\mathcal{O}(FKT)$	50	$1\times$
Convolutive NMF	$\mathcal{O}(PFKT)$	5	$\sim 0.5\times$

The relative total cost column reflects the combined effect of iteration count and per-iteration complexity. All methods were run within a comparable total wall-time budget per track ($<$30 s on a commodity CPU), enabling a fair time-controlled comparison.

5 Conclusion

We presented a controlled empirical study of four NMF formulations for supervised music source separation: Euclidean NMF, Itakura-Saito NMF, Sparse NMF, and Convolutive NMF. All methods were trained on 20 MUSDB18 training tracks and evaluated on 25 valid test tracks under a fixed wall-time budget, using BSS Eval metrics.

The main findings are as follows. First, all four methods yield negative SDR for vocals, bass, and drums, confirming that NMF with $K_s = 12$ components per source is severely under-specified for real-world mixtures. The "other" accompaniment track is the only source where positive SDR is consistently achieved. Second, Sparse NMF is the most consistently effective variant, achieving the best score on 8 of 12 source/metric combinations. The ℓ_1 penalty reduces cross-source bleed and reconstruction artifacts with negligible computational overhead over the Euclidean baseline. Third, IS-NMF provides a meaningful improvement for tonal, melodic sources (vocal SDR $+2.6$ dB), confirming that the scale-invariant IS divergence is better matched to the log-amplitude statistics of pitched audio. Fourth, CNMF's performance on drums degrades under the tight iteration budget imposed by its high per-iteration cost. Given sufficient compute, its multi-frame patches should yield improvements for transient sources.

A few practical takeaways follow from these results. Sparse NMF is the recommended NMF variant when separation quality is the primary objective and compute is limited. IS-NMF is preferable when tonal sources are the focus and a modest increase in per-iteration cost is acceptable. CNMF requires a substantially larger iteration budget to realize its expressive advantage. Future CPU implementations should exploit vectorized shift operations to reduce overhead.

Future work should explore combining the IS divergence with the ℓ_1 sparsity penalty, which may capture both the perceptual and structural benefits simultaneously. Increasing K_s and using a longer audio excerpt or the full track are straightforward extensions that are expected to close the gap with deep-learning baselines within the interpretable NMF framework. Incorporating harmonic constraints [12] as a structured prior on $\mathbf{W}$ also represents a promising direction.

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